



Lecture 31

Least Squares

Announcements

- **Project 2 checkpoint due today at 5 pm!**
 - Final deadline on April 12th
 - Homework 10 due 04/10
 - Chapter 15 of the textbook is helpful!
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Regression Roadmap

- Monday
 - How to measure linear association
 - Wednesday
 - Predicting one numerical variable from a another
 - The regression line
 - **Today**
 - The “best” linear predictor
 - The method of least squares
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Linear Regression

Prediction

Goal: Predict y using x

Examples:

- Predict # *hospital beds available* using *air pollution*
- Predict *house prices* using *house size*
- Predict # *app users* using # *app downloads*

So far we have been using numerical variables but it is possible to use categorical variables in the regression as well

Regression Estimate

Goal: Predict y using x

To find the regression estimate of y :

- Convert the given x to standard units
 - Multiply by r
 - That's the regression estimate of y , but:
 - It's in standard units
 - So convert it back to the original units of y
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Slope and Intercept (in original units)

estimate of y = slope * x + intercept

$$\text{slope of the regression line} = r \cdot \frac{\text{SD of } y}{\text{SD of } x}$$

$$\text{intercept of the regression line} = \text{average of } y - \text{slope} \cdot \text{average of } x$$

Discussion Question

A course has a midterm (average 70; standard deviation 10) and a really hard final (average 50; standard deviation 12)

If the scatter diagram comparing midterm & final scores for students has an oval shape with correlation 0.75, then...

What do you expect the average final score would be for students who scored 60 on the midterm?

Discussion Question

Suppose we use linear regression to predict candy prices (in dollars) from sugar content (in grams). What are the units of each of the following?

- r
 - The slope
 - The intercept
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Least Squares

(Demo)

Error in Estimation

- **error = actual value – estimate**
- Typically, some errors are positive and some negative
- To measure the rough size of the errors
 - **square** the **errors** to eliminate cancellation
 - take the **mean** of the squared errors
 - take the square **root** to fix the units
 - **root mean square error** (rmse)

(Demo)

Numerical Optimization

- Numerical minimization is approximate but effective
- Lots of machine learning uses numerical minimization
- If the function `mse(a, b)` returns the mse of estimation using the line “estimate = $ax + b$ ”,
 - then `minimize(mse)` returns array `[a0, b0]`
 - `a0` is the slope and `b0` the intercept of the line that *minimizes* the mse among lines with arbitrary slope `a` and arbitrary intercept `b` (that is, among all lines)

(Demo)

Least Squares Line

- Minimizes the root mean squared error (rmse) among all lines
 - Equivalently, minimizes the mean squared error (mse) among all lines
 - Minimizing mean squared error gives us the regression line
 - Regression line is the only estimate that minimizes the MSE.
 - Names:
 - “Best fit” line
 - Least squares line
 - Regression line
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